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**EDUCATION**

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| <b>The Chinese University of Hong Kong (CUHK)</b>                                                                 | Hong Kong SAR |
| <i>Doctor of Philosophy , Geophysics (Seismology)</i>                                                             | 2026 – 2030   |
| <b>University of Tehran, Institute of Geophysics</b>                                                              | Tehran, Iran  |
| <i>Master of Science, Geophysics (Exploration Seismology)</i>                                                     | 2025 – 2026   |
| • Withdrew to accept Ph.D. fellowship                                                                             |               |
| <b>Kharazmi University, Faculty of Earth Sciences</b>                                                             | Tehran, Iran  |
| <i>Bachelor of Science, Geology</i>                                                                               | 2021 – 2025   |
| • Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025)                                  |               |
| • Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024)                  |               |
| <b>University of Lille, International Academy Lille HdF</b>                                                       | Lille, France |
| <i>Water Resource Management Summer School (5 ECTS)</i>                                                           | Summer 2024   |
| • Presented a numerical groundwater modeling study on water abstraction impacts in the Weyib Watershed, Ethiopia. |               |

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**TEACHING EXPERIENCE**

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| <b>Iranian Young Scholars Club (IrYSC)</b>                                                         | Tehran, Iran              |
| <i>Earth Science Olympiad Teacher</i>                                                              | April 2021 – August 2026  |
| <b>Kharazmi University</b>                                                                         | Tehran, Iran              |
| <i>Programming Teaching Assistant</i>                                                              | April 2025 – June 2025    |
| • Programming Course, under supervision of Dr. Morteza Asemani                                     |                           |
| <b>Kharazmi University</b>                                                                         | Tehran, Iran              |
| <i>Geostatistics Teaching Assistant</i>                                                            | February 2025 – June 2025 |
| • Geostatistics Course, under supervision of Dr. Ehsan Pegah                                       |                           |
| <b>Young Scholars Club</b>                                                                         | Tehran, Iran              |
| <i>Instructor</i>                                                                                  | July 2024                 |
| • Taught "Applied Data Science in Geology" for the Iran National IESO <sup>1</sup> 2024 Team       |                           |
| <b>Kharazmi University</b>                                                                         | Tehran, Iran              |
| <i>Instructor</i>                                                                                  | April 2024 – June 2024    |
| • Organized and instructed the Advanced Python and Machine Learning Workshop                       |                           |
| <b>NODET<sup>2</sup></b>                                                                           | Tehran, Iran              |
| <i>Teacher</i>                                                                                     | May 2021 – September 2021 |
| • Taught ITEO <sup>3</sup> Marketing at Allameh Helli (Branches 5 & 8) and Farzanegan High Schools |                           |

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**RESEARCH EXPERIENCE**

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| <b>FEZtool</b>                                                                                        | Tehran, Iran                |
| <i>Co-Founder &amp; Research and Development Specialist</i>                                           | August 2023 – Present       |
| • Led Earth science research using advanced algorithms and geospatial analysis.                       |                             |
| <b>International Astronomical Search Collaboration, NASA Partner</b>                                  | Remote                      |
| <i>Asteroid Search Team Member</i>                                                                    | July 2025 – September 2025  |
| • Analyzed Pan-STARRS data, contributing to 3 preliminary asteroid discoveries.                       |                             |
| <b>Kharazmi University</b>                                                                            | Tehran, Iran                |
| <i>Research Assistant</i>                                                                             | October 2024 – January 2025 |
| • Performed MASW <sup>4</sup> to extract dispersion images under supervision of Dr. Mehdi Talkhablou. |                             |

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**COMMUNITY SERVICE**

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|-------------------------------------------------------------------------|-------------------------|
| <b>Zamin Pouya Scientific Journal Vol. 31</b>                           | Tehran, Iran            |
| <i>Associate Editor &amp; Reviewer</i>                                  | August 2025 – June 2026 |
| <b>The 4<sup>th</sup> National Festival of Young Iranian Geologists</b> | Tehran, Iran            |
| <i>Education Committee Member</i>                                       | April 2025 – May 2025   |

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<sup>1</sup>International Earth Science Olympiad<sup>2</sup>National Organization for Development of Exceptional Talents<sup>3</sup>Iranian Thinking & Entrepreneurship Olympiad<sup>4</sup>Multichannel Analysis of Surface Waves

SELECTED PUBLICATIONS

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## Journal Articles

1. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. *Kharazmi Journal of Earth Sciences*.
2. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. *Ovan Scientific Journal*
3. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashef, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. *Journal of Water Management Modeling (JWMM)*

## Conference Papers

1. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In *Seismology Student Workshop (SSW 2025)*, GatherTown (Virtual)
2. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In *The 1<sup>st</sup> Conference on Applied Geophysics in Mines*, Tehran, Iran
3. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In *2<sup>nd</sup> International Conference for Iranian Geography and Earth Sciences Students*, Tehran, Iran
4. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In *8<sup>th</sup> Symposium of Sedimentological Society of Iran*, Hormozgan, Iran
5. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In *26<sup>th</sup> Conference of Geological Society of Iran*, Urmia, Iran

## Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence*. **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning*. **Tasir Publication**. (Original work published by Maurizio Petrelli)

PROJECTS

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**PlanetaryGeophysics** | [github.com/aradfarahani/PlanetaryGeophysics](https://github.com/aradfarahani/PlanetaryGeophysics) 2025 – Present

- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.

**FEZrs** (Open Source) | [pypi.org/project/fezrs](https://pypi.org/project/fezrs) 2023 – Present

- Advanced Python library for remote sensing, developed by FEZtool with over 100,000 downloads

**Seismicity Analysis in Iran (2010–2025)** | [iris.edu/hq/workshops/2025/06](https://iris.edu/hq/workshops/2025/06) 2025

- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

**Goelectricspy** (Open Source) | [zenodo.org/records/10966815](https://zenodo.org/records/10966815) 2023

- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

INVITED TALKS

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1. **Walrus: A Cross-Domain Foundation Model for Continuum Dynamics:** *Applied ML Lab, Center for Complex Networks, Department of Physics, Shahid Beheshti University*; 13 December 2025

2. **Applied Artificial Intelligence in Earth Science:** *Faculty of Geosciences, Shahrood University of Technology & Faculty of Earth Sciences, Kharazmi University*; 11 November 2025
3. **Applications of Artificial Intelligence in Education and Earth Science Analysis:** *Educational Technology Expert Group, Karaj and Fardis County Department of Education, District 2*; 30 December 2024

**LICENSES & CERTIFICATIONS**

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|---------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Applied Geotechnical Seismology</b> , Kharazmi University                                                                                | 2026 |
| <b>Applications of Non-Destructive Geophysical Methods in Engineering Geology Studies</b> , Kharazmi University                             |      |
| <b>Seismology Skill Building Workshop 2025</b> , EarthScope Consortium                                                                      | 2025 |
| <b>Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method</b> , National Iranian Geophysical Society |      |
| <b>Remote Pilot Certification (Multirotor)</b> , Civil Aviation Authority of Islamic Republic of Iran                                       | 2023 |
| <b>Unmanned Aerial Vehicle Surveying</b> , Maan Academy                                                                                     |      |
| <b>GIS, Mapping, and Spatial Analysis Specialization</b> , University of Toronto (Offered by Coursera)                                      |      |
| <b>CS50 Certificate</b> , CS50 (Harvard University)                                                                                         | 2021 |

**HONORS & AWARDS**

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| <b>CUHK’s International Student PhD Fellowship - Awardee</b> , CUHK                              | 2026 |
| <b>9<sup>th</sup> National Young Scientists Festival - Merit Award Winner</b> , Geology Category | 2025 |
| <b>Hydro 2024 Research-Oriented Summer School - Full Scholarship</b> , University of Lille       | 2024 |
| <b>Research Visit Grant (2500 DKK) - Awardee</b> , University of Copenhagen                      | 2022 |
| <b>DrCT 2021 Global Computational Competition - Gold Medalist</b>                                | 2021 |
| <b>1<sup>st</sup> national Thinking &amp; Entrepreneurship Olympiad (ITEO) - Silver Medalist</b> | 2020 |
| <b>WMI 2015 World Mathematical Championships Selection - Silver Medalist</b>                     | 2015 |

**SKILLS & INTERESTS**

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| <b>Programming Languages:</b> Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R |  |
| <b>Markup Languages:</b> Markdown, L <sup>A</sup> T <sub>E</sub> X                                                                                                                                                                       |  |
| <b>Softwares:</b> Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas                                                         |  |
| <b>Field Techniques:</b> Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition                                                                                  |  |
| <b>Languages:</b> Farsi (Persian): <i>Native</i> ; English: <i>Fluent (C1)</i> ; German: <i>Elementary (A2)</i> ; Mandarin: <i>Beginner (Duolingo Score: 10)</i>                                                                         |  |
| <b>Research Interests:</b> Signal/Image Processing, Geophysics (Seismology), Remote Sensing, Machine Learning, Open-Source Software Development                                                                                          |  |

**PROFESSIONAL MEMBERSHIPS**

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|---------------------------------------------|--------------------------------|
| <b>National Iranian Geophysical Society</b> | Student member, 2025 – Present |
| <b>Space Generation Advisory Council</b>    | Member, 2025 – Present         |
| <b>Google Maps Platform Innovators</b>      | Member, 2025 – Present         |
| <b>Geological Society of Iran</b>           | Student member, 2021 – 2022    |